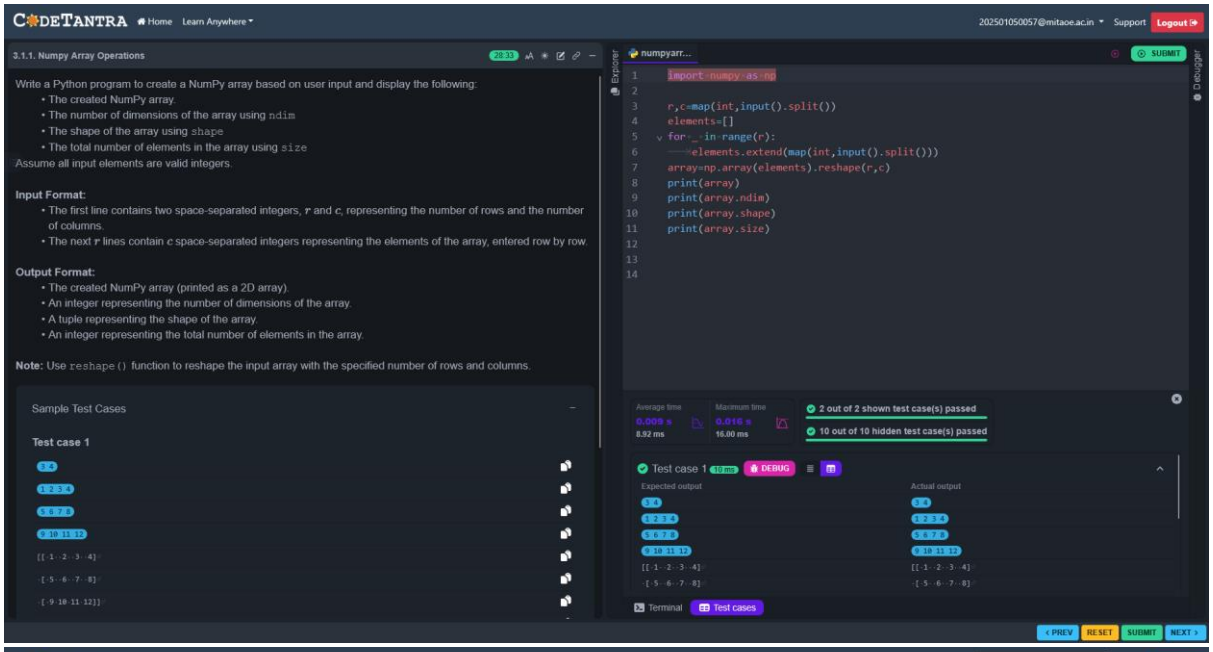
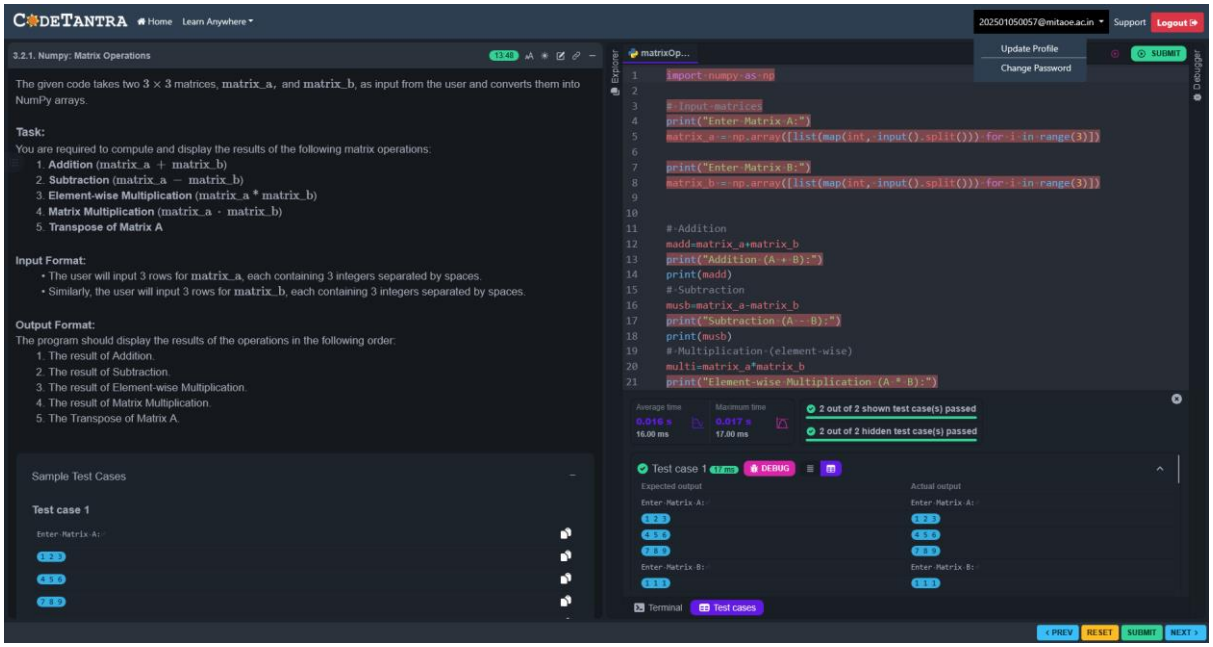
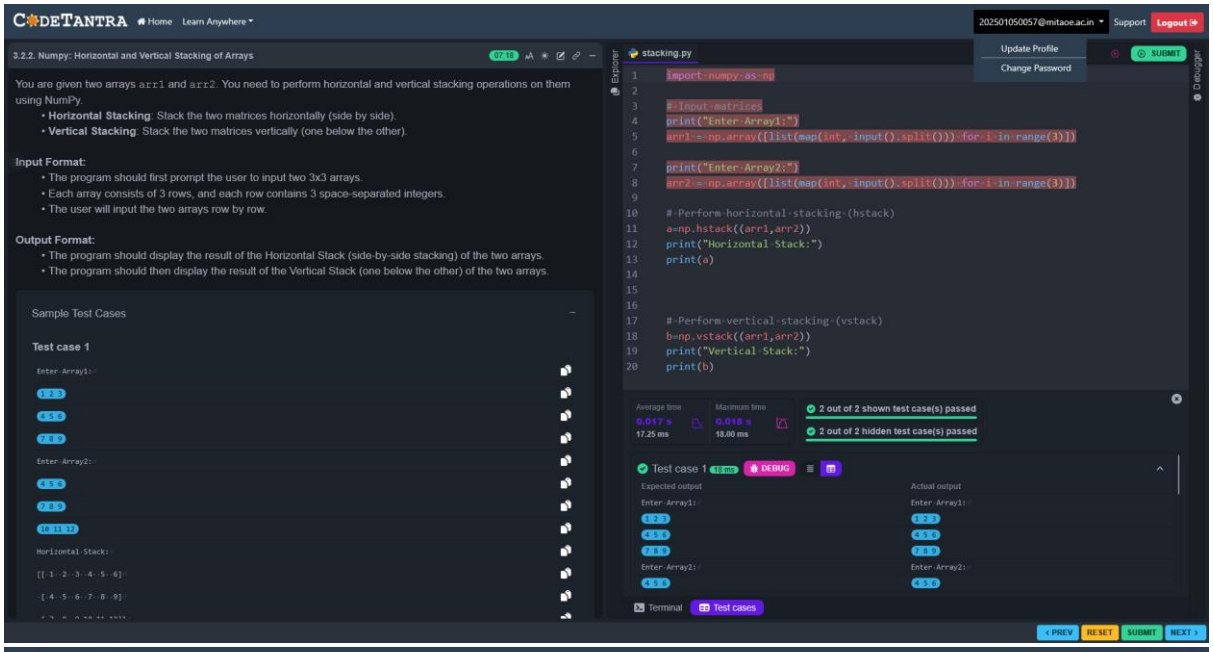
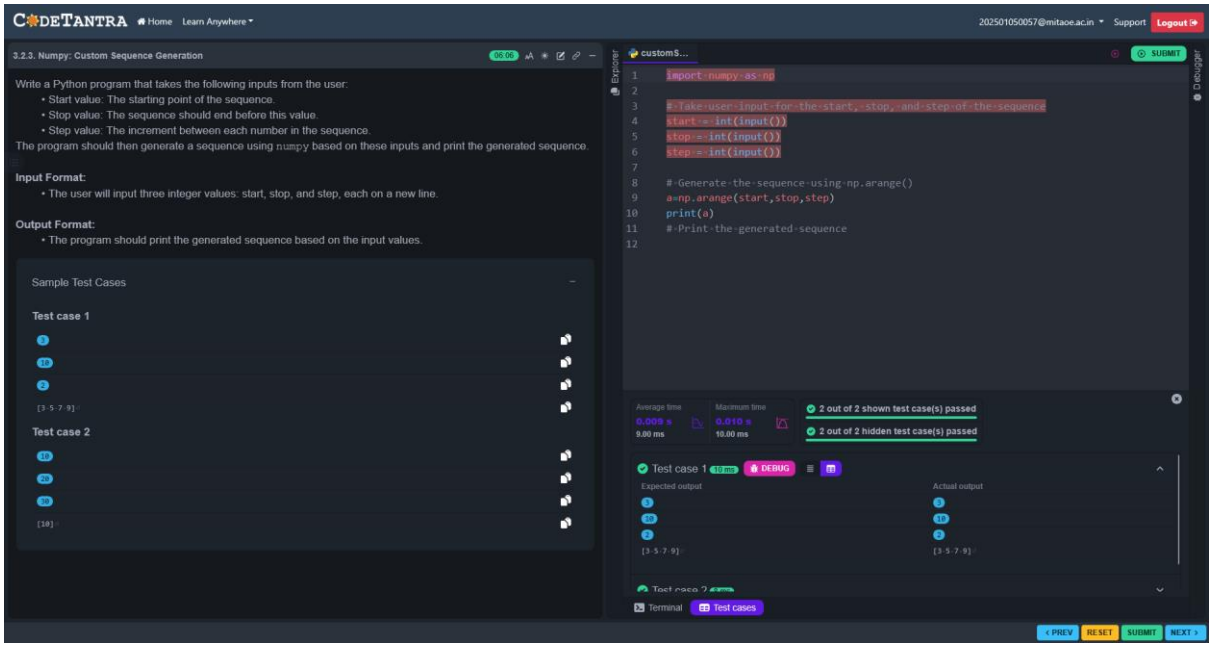
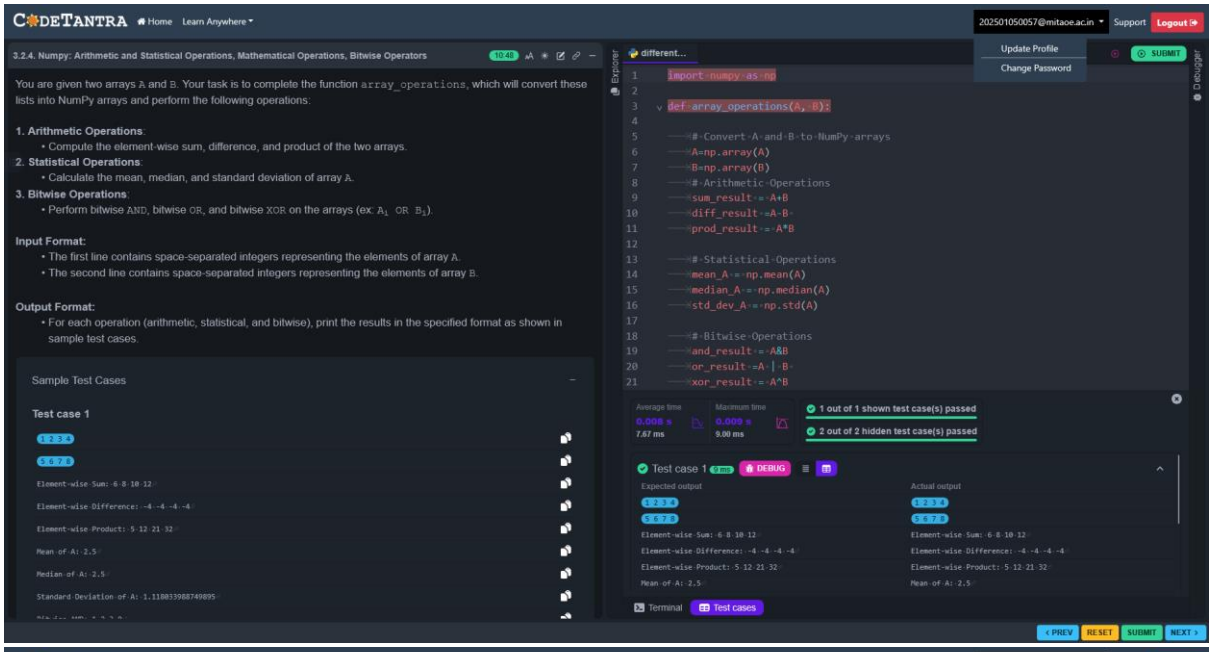
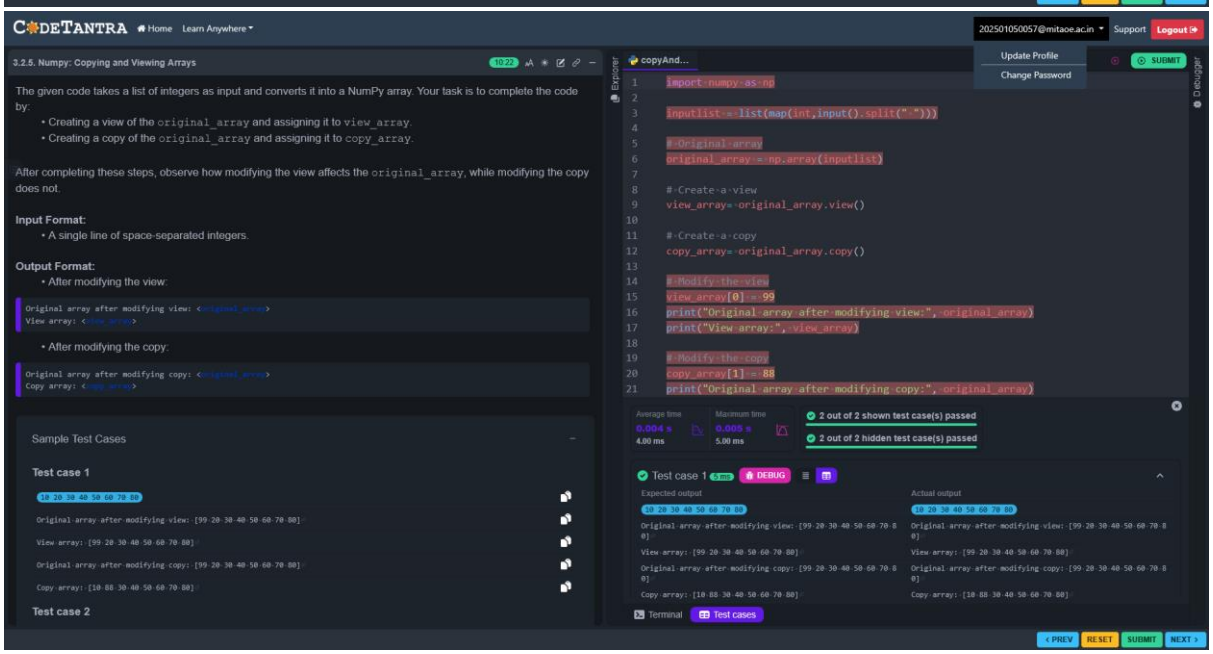

CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
3.1.1. Numpy Array Operations
Write a Python program to create a NumPy array based on user input and display the following:
• The created NumPy array.
• The number of dimensions of the array using ndim
• The shape of the array using shape
• The total number of elements in the array using size
Assume all input elements are valid integers.
Input Format:
• The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
• The next r lines contain c space-separated integers representing the elements of the array, entered row by row.
Output Format:
• The created NumPy array (printed as a 2D array).
• An integer representing the number of dimensions of the array.
• A tuple representing the shape of the array.
• An integer representing the total number of elements in the array.
Note: Use reshape() function to reshape the input array with the specified number of rows and columns.
Sample Test Cases
Test case 1
3 4
1 2 3 4
5 6 7 8
9 10 11 12
[[ 1 2 3 4]
[ 5 6 7 8]
[ 9 10 11 12]]
numpyarr...
SUBMIT
1 import numpy as np
2
3 r,c=map(int,input().split())
4 elements=[]
5 for _ in range(r):
6 elements.extend(map(int,input().split()))
7 array=np.array(elements).reshape(r,c)
8 print(array)
9 print(array.ndim)
10 print(array.shape)
11 print(array.size)
12
13
14
Average time
0.009 s
8.92 ms
Maximum time
0.016 s
16.00 ms
2 out of 2 shown test case(s) passed
10 out of 10 hidden test case(s) passed
Test case 1
DEBUG
Expected output
Actual output
Terminal
Test cases
‹ PREV
RESET
SUBMIT
NEXT ›

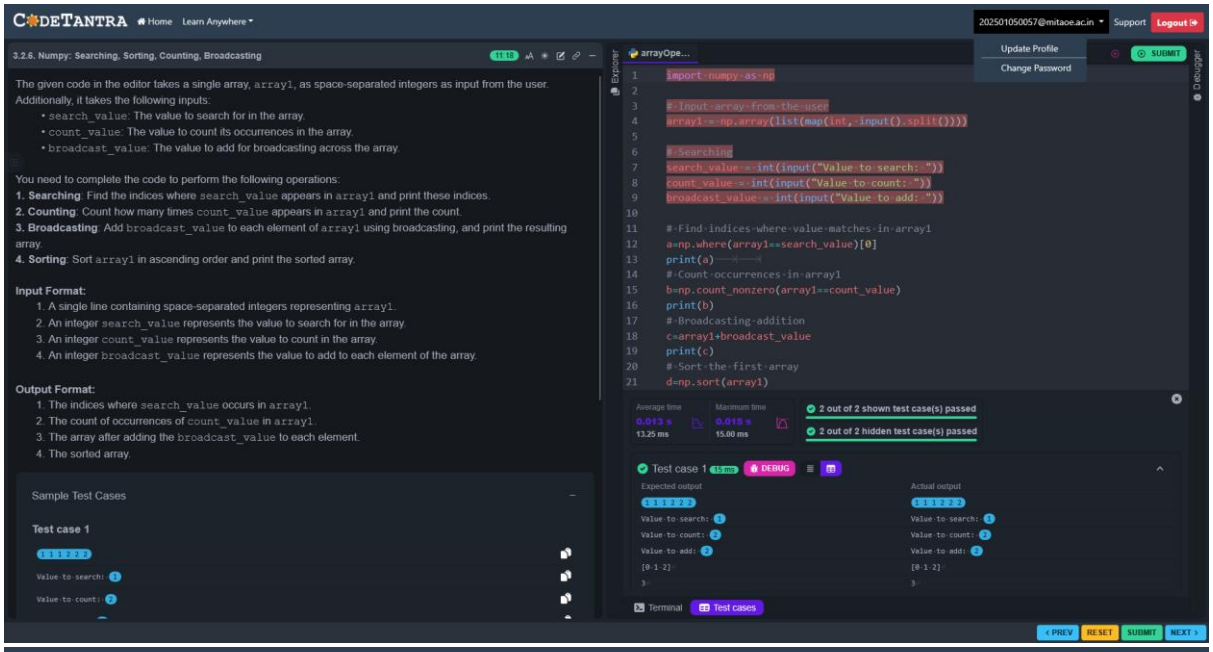

CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.1. Numpy: Matrix Operations
The given code takes two 3 × 3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.
Task:
You are required to compute and display the results of the following matrix operations:
1. Addition (matrix_a + matrix_b)
2. Subtraction (matrix_a − matrix_b)
3. Element-wise Multiplication (matrix_a * matrix_b)
4. Matrix Multiplication (matrix_a · matrix_b)
5. Transpose of Matrix A
Input Format:
• The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
• Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.
Output Format:
The program should display the results of the operations in the following order:
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.
Sample Test Cases
Test case 1
Enter Matrix A:
1 2 3
4 5 6
7 8 9
matrixOp...
SUBMIT
1 import numpy as np
2
3 # Input matrices
4 print("Enter Matrix A:")
5 matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Matrix B:")
8 matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
9
10
11 # Addition
12 madd=matrix_a+matrix_b
13 print("Addition (A + B):")
14 print(madd)
15 # Subtraction
16 musb=matrix_a-matrix_b
17 print("Subtraction (A - B):")
18 print(musb)
19 # Multiplication (element-wise)
20 multi=matrix_a*matrix_b
21 print("Element-wise Multiplication (A * B):")
Average time
0.016 s
16.00 ms
Maximum time
0.017 s
17.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
DEBUG
Expected output
Actual output
Terminal
Test cases
‹ PREV
RESET
SUBMIT
NEXT ›

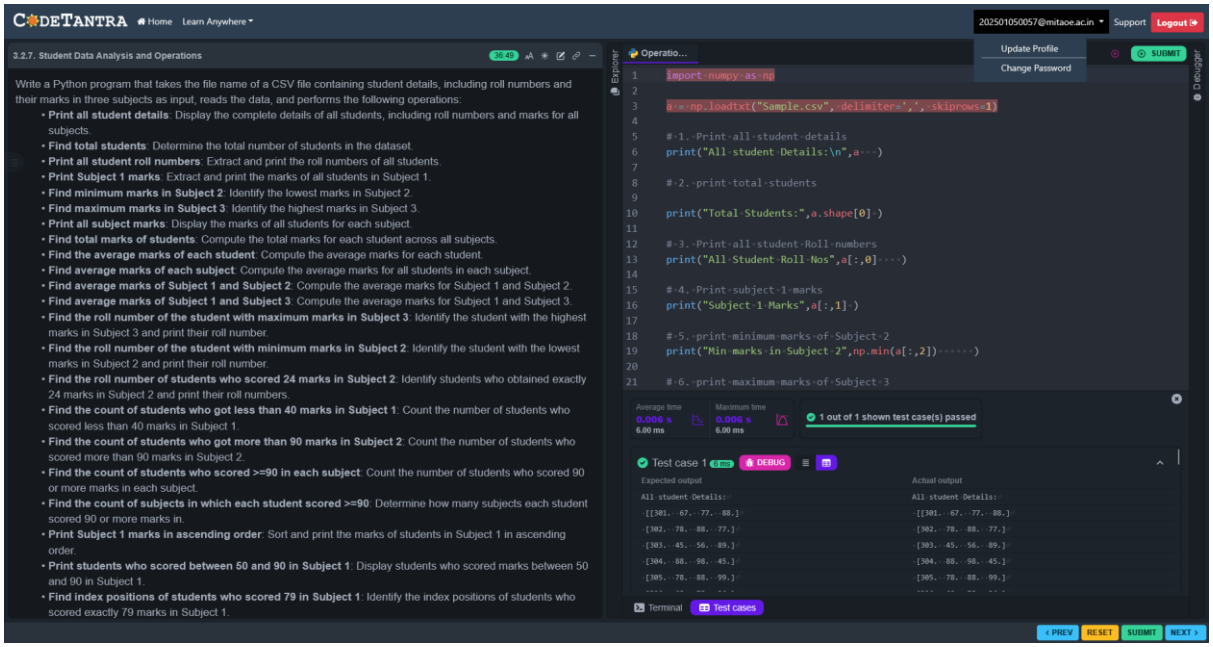

CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays
07:11
You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.
• Horizontal Stacking: Stack the two matrices horizontally (side by side).
• Vertical Stacking: Stack the two matrices vertically (one below the other).
Input Format:
• The program should first prompt the user to input two 3x3 arrays.
• Each array consists of 3 rows, and each row contains 3 space-separated integers.
• The user will input the two arrays row by row.
Output Format:
• The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
• The program should then display the result of the Vertical Stack (one below the other) of the two arrays.
Sample Test Cases
Test case 1
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[ 1 2 3 4 5 6]
[ 4 5 6 7 8 9]
stacking.py
SUBMIT
Debugger
Explorer
import numpy as np
#Input matrices
print("Enter Array1:")
arr1=np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Array2:")
arr2=np.array([list(map(int, input().split())) for i in range(3)])
#Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)
#Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
Average time
0.017 s
17.25 ms
Maximum time
0.018 s
18.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
18 ms
DEBUG
Expected output
Actual output
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT


CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
3.2.3. Numpy: Custom Sequence Generation
06:06
Write a Python program that takes the following inputs from the user:
• Start value: The starting point of the sequence.
• Stop value: The sequence should end before this value.
• Step value: The increment between each number in the sequence.
The program should then generate a sequence using numpy based on these inputs and print the generated sequence.
Input Format:
• The user will input three integer values: start, stop, and step, each on a new line.
Output Format:
• The program should print the generated sequence based on the input values.
Sample Test Cases
Test case 1
3
10
2
[3 5 7 9]
Test case 2
10
20
30
[10]
customS...
SUBMIT
Debugger
Explorer
import numpy as np
#Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
#Generate the sequence using np.arange()
a=np.arange(start,stop,step)
print(a)
#Print the generated sequence
Average time
0.009 s
9.00 ms
Maximum time
0.010 s
10.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
18 ms
DEBUG
Expected output
Actual output
3
10
2
[3 5 7 9]
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT


CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators
10:48
You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:
1. Arithmetic Operations:
• Compute the element-wise sum, difference, and product of the two arrays.
2. Statistical Operations:
• Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations:
• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A1 OR B1).
Input Format:
• The first line contains space-separated integers representing the elements of array A.
• The second line contains space-separated integers representing the elements of array B.
Output Format:
• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.
Sample Test Cases
Test case 1
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Median of A: 2.5
Standard Deviation of A: 1.118033988749895
different...
SUBMIT
Debugger
Explorer
1 import numpy as np
2
3 def array_operations(A, B):
4
5 #Convert A and B to NumPy arrays
6 A=np.array(A)
7 B=np.array(B)
8 #Arithmetic Operations
9 sum_result = A+B
10 diff_result =A-B
11 prod_result = A*B
12
13 #Statistical Operations
14 mean_A = np.mean(A)
15 median_A = np.median(A)
16 std_dev_A = np.std(A)
17
18 #Bitwise Operations
19 and_result = A&B
20 or_result =A | B
21 xor_result = A^B
Average time
0.008 s
7.67 ms
Maximum time
0.009 s
9.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
3 ms
DEBUG
Expected output
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Actual output
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT


CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.5. Numpy: Copying and Viewing Arrays
10:22
The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:
• Creating a view of the original_array and assigning it to view_array.
• Creating a copy of the original_array and assigning it to copy_array.
After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.
Input Format:
• A single line of space-separated integers.
Output Format:
• After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>
• After modifying the copy:
Original array after modifying copy: <original_array>
Copy array: <copy_array>
Sample Test Cases
Test case 1
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Test case 2
copyAnd...
SUBMIT
Debugger
Explorer
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 #Original array
6 original_array = np.array(inputlist)
7
8 #Create a view
9 view_array= original_array.view()
10
11 #Create a copy
12 copy_array= original_array.copy()
13
14 #Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
18
19 #Modify the copy
20 copy_array[1] = 88
21 print("Original array after modifying copy:", original_array)
Average time
0.004 s
4.00 ms
Maximum time
0.005 s
5.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
3 ms
DEBUG
Expected output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Actual output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT


CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting
11:11
SUBMIT
The given code in the editor takes a single array, array1, as space-separated integers as input from the user. Additionally, it takes the following inputs:
• search_value: The value to search for in the array.
• count_value: The value to count its occurrences in the array.
• broadcast_value: The value to add for broadcasting across the array.
You need to complete the code to perform the following operations:
1. Searching: Find the indices where search_value appears in array1 and print these indices.
2. Counting: Count how many times count_value appears in array1 and print the count.
3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. Sorting: Sort array1 in ascending order and print the sorted array.
Input Format:
1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.
Output Format:
1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.
Sample Test Cases
Test case 1
1 1 1 2 2 2
Value to search: 1
Value to count: 2
arrayOpe...
import numpy as np
# Input array from the user
array1 = np.array(list(map(int, input().split())))
# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)
# Broadcasting addition
c=array1+broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
Average time
0.043 s
13.25 ms
Maximum time
0.018 s
15.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
DEBUG
Expected output
Actual output
Terminal
Test cases
‹ PREV
RESET
SUBMIT
NEXT ›


CODETANTRA
Home
Learn Anywhere
202501050057@mitaoe.ac.in
Support
Logout
Update Profile
Change Password
3.2.7. Student Data Analysis and Operations
36:49
SUBMIT
Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:
• Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
• Find total students: Determine the total number of students in the dataset.
• Print all student roll numbers: Extract and print the roll numbers of all students.
• Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
• Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
• Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
• Print all subject marks: Display the marks of all students for each subject.
• Find total marks of students: Compute the total marks for each student across all subjects.
• Find the average marks of each student: Compute the average marks for each student.
• Find average marks of each subject: Compute the average marks for all students in each subject.
• Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
• Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
• Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
• Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
• Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
• Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
• Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
• Find the count of students who scored >=90 in each subject: Count the number of students who scored 90 or more marks in each subject.
• Find the count of subjects in which each student scored >=90: Determine how many subjects each student scored 90 or more marks in.
• Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
• Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
• Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.
Operatio...
import numpy as np
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
#1. Print all student details
print("All student Details:\n",a)
#2. print total students
print("Total Students:",a.shape[0])
#3. Print all student Roll numbers
print("All Student Roll Nos",a[:,0])
#4. Print subject 1 marks
print("Subject 1 Marks",a[:,1])
#5. print minimum marks of Subject 2
print("Min marks in Subject 2",np.min(a[:,2]))
#6. print maximum marks of Subject 3
Average time
0.006 s
6.00 ms
Maximum time
0.006 s
6.00 ms
1 out of 1 shown test case(s) passed
Test case 1
DEBUG
Expected output
Actual output
All student Details:
[[301. 67. 77. 88.]
[302. 78. 88. 77.]
[303. 45. 56. 89.]
[304. 88. 98. 45.]
[305. 78. 88. 99.]
Terminal
Test cases
‹ PREV
RESET
SUBMIT
NEXT ›